

Independent Validation Report

Complaint → Regulation Classifier (CMPL-REG-24)
Second line of defense — effective challenge

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LLM: nim (meta/llama-3.1-8b-instruct)
Data: CFPB Consumer Complaint Database (real, redacted narratives)
Dataset: 4000 rows · regulatory rate 0.9663
Repo: reg-agents · model id: CMPL-REG-24

1 • Assessment narrative

The two-stage complaint classification model, comprising stage-1 TF-IDF classifiers and stage-2 retrieval-augmented generation pipeline (rag_llm), has been thoroughly examined in this validation report. The model's conceptual soundness, data quality, outcomes analysis, monitoring and guardrails adequacy, and consumer-protection/fair-lending exposure of complaint routing have been scrutinized.

Conceptual soundness is a critical aspect of any model, and the two-stage design employed here is a common approach in complaint classification tasks. However, it is essential to maintain the assumptions underlying this design. The stage-1 classifiers are expected to provide a robust feature representation, which is then refined by the stage-2 rag_llm pipeline. The assumption that the stage-1 classifiers are effective in capturing the relevant features is critical to the overall performance of the model. If this assumption is not met, the stage-2 pipeline may not be able to improve upon the stage-1 performance.

The data quality is a significant concern, given the weak-supervision reference used to derive the labels. The CFPB product/issue taxonomy and narrative keyword rules are used to assign labels to the complaint narratives. While this approach is common in the absence of human-adjudicated labels, it introduces a level of uncertainty and potential bias in the labels. This, in turn, affects every downstream metric, including the distinction between measured agreement and true accuracy. The validation report flags this as a limitation, and it is essential to acknowledge the potential impact on the model's performance.

The outcomes analysis reveals a mixed picture. The stage-1 TF-IDF classifiers perform well, with the logistic_regression model achieving a pr_auc of 0.9962 and an accuracy of 0.95. However, the stage-2 rag_llm pipeline struggles, with an accuracy of 0.3652 and a family_accuracy of 0.5478. The recall values for individual labels are also concerning, with several labels having a recall of 0.0. This suggests that the stage-2 pipeline may not be effective in capturing the nuances of the complaint narratives.

The monitoring and guardrails adequacy are also critical aspects of the model's performance. The model's ability to detect and respond to changes in the complaint landscape is essential to its effectiveness. However, the validation report does not provide sufficient information to assess the adequacy of the monitoring and guardrails.

The consumer-protection/fair-lending exposure of complaint routing is a significant concern. The model's ability to route complaints to the relevant regulatory agencies is critical to ensuring that consumers receive the necessary protection. However, the validation report does not provide sufficient information to assess the model's performance in this regard.

In light of these findings, the following validation findings are presented:

V-1: ****High**** - The model's conceptual soundness is compromised by the assumption that the stage-1 classifiers are effective in capturing the relevant features. Remediation: Re-evaluate the stage-1 classifiers and refine the feature representation to improve the stage-2 pipeline's performance. Owner: Model Development Team.

V-2: ****High**** - The data quality is compromised by the weak-supervision reference used to derive the labels. Remediation: Consider using human-adjudicated labels or alternative labeling approaches to improve the label quality. Owner: Data Science Team.

V-3: ****Medium**** - The stage-2 rag_llm pipeline struggles to achieve high accuracy and recall values. Remediation: Refine the rag_llm pipeline and explore alternative approaches to improve its performance. Owner: Model Development Team.

1 - Assessment narrative (cont.)

V-4: **Medium** - The monitoring and guardrails adequacy are not sufficient to assess the model's performance. Remediation: Develop and implement a robust monitoring and guardrails framework to ensure the model's effectiveness. Owner: Model Operations Team.

V-5: **High** - The consumer-protection/fair-lending exposure of complaint routing is not adequately addressed. Remediation: Develop and implement a robust complaint routing framework to ensure that consumers receive the necessary protection. Owner: Model Operations Team.

In light of these findings, the disposition of the validation report is:

Approve with Conditions

Conditions:

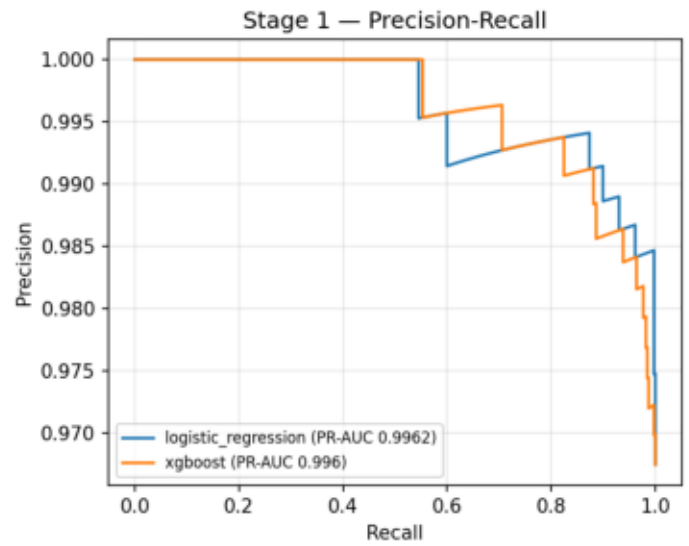
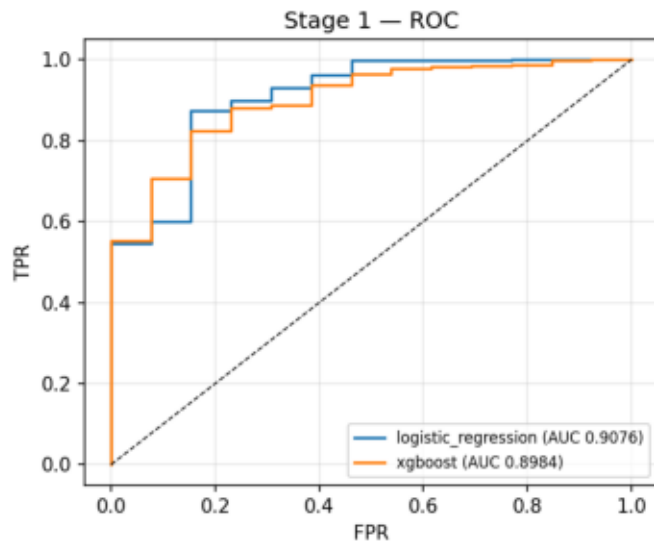
1. The model development team must re-evaluate the stage-1 classifiers and refine the feature representation to improve the stage-2 pipeline's performance.
2. The data science team must consider using human-adjudicated labels or alternative labeling approaches to improve the label quality.
3. The model development team must refine the rag_llm pipeline and explore alternative approaches to improve its performance.
4. The model operations team must develop and implement a robust monitoring and guardrails framework to ensure the model's effectiveness.
5. The model operations team must develop and implement a robust complaint routing framework to ensure that consumers receive the necessary protection.

These conditions must be met before the model can be deployed in production.

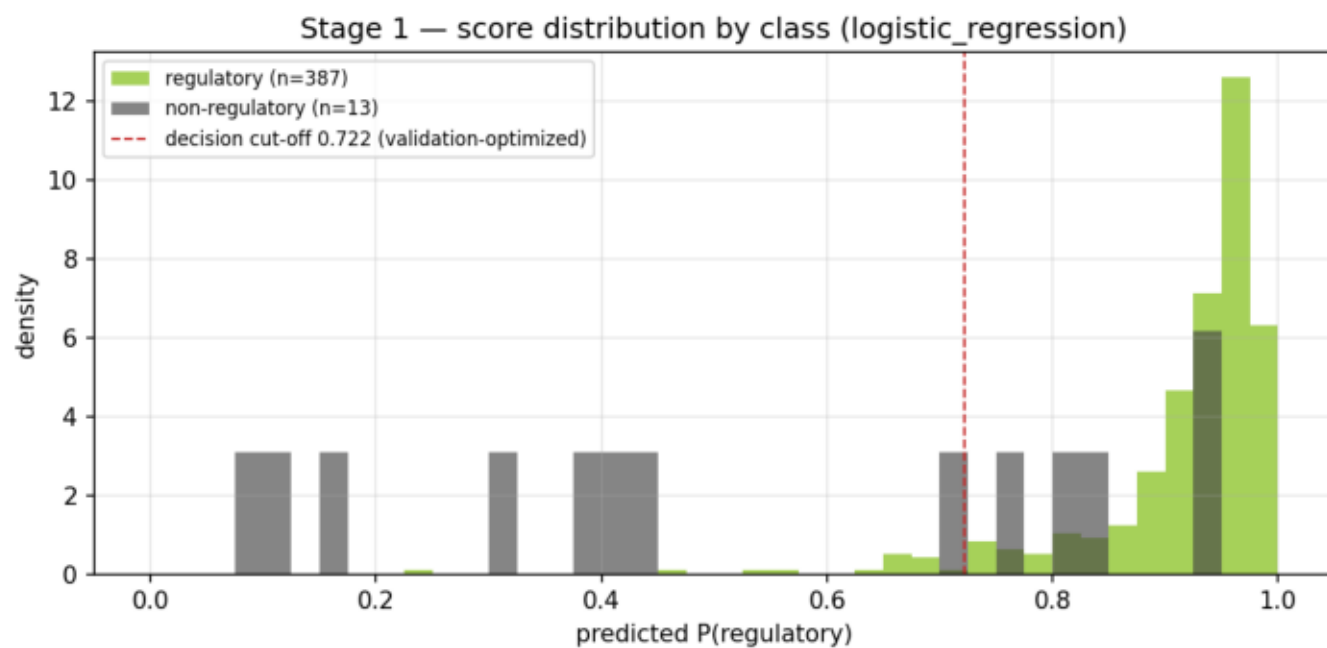
2 · Stage-1 leaderboard (validated re-run)

model	val_pr_auc	threshold	pr_auc	roc_auc	f1	precision	recall	accuracy
logistic_regress	0.9958	0.722	0.9962	0.9076	0.9738	0.9867	0.9612	0.95
xgboost	0.9938	0.585	0.996	0.8984	0.9766	0.9817	0.9716	0.955

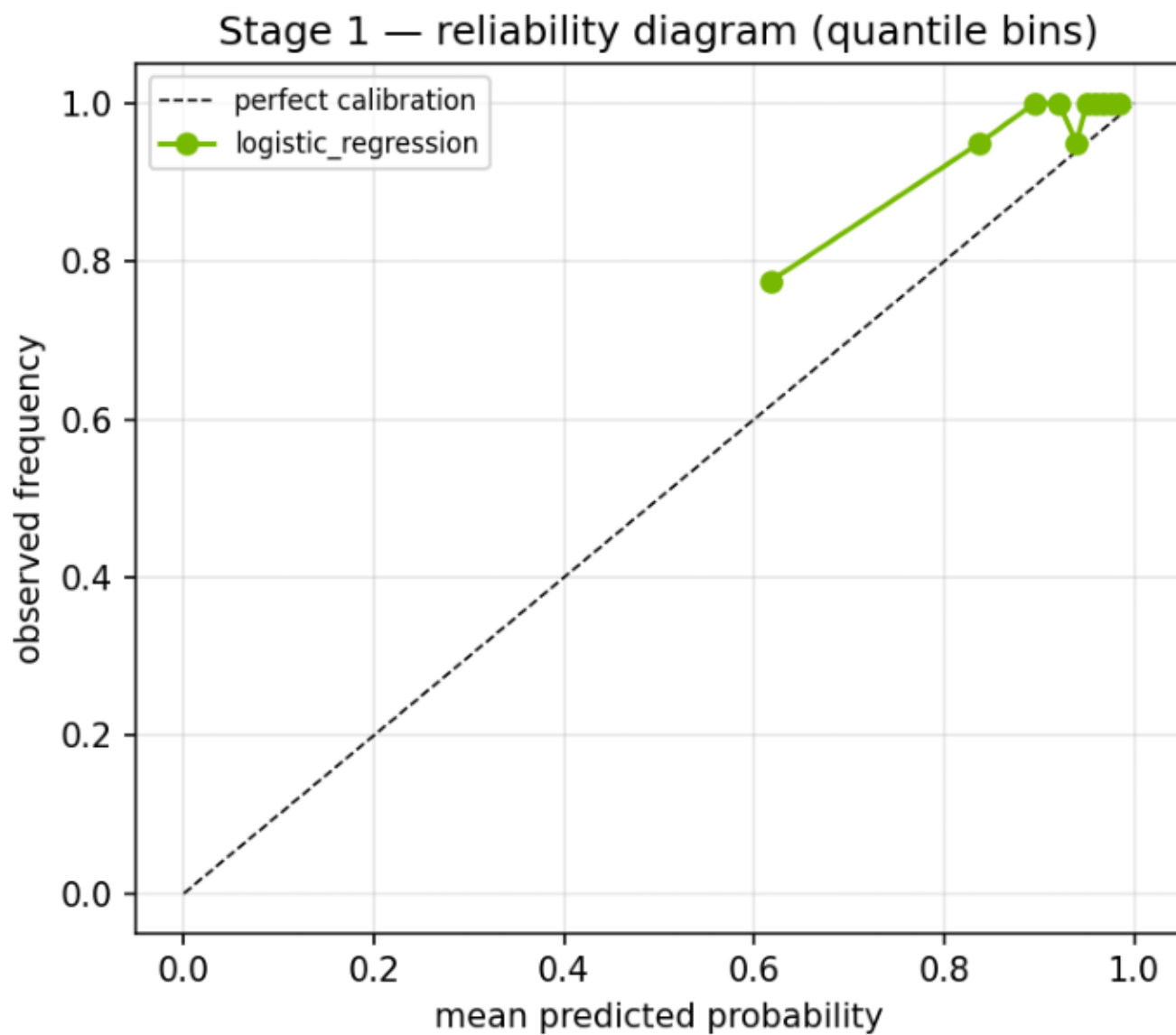
2 · Stage-1 evidence: discrimination



2 · Stage-1 evidence: score separation

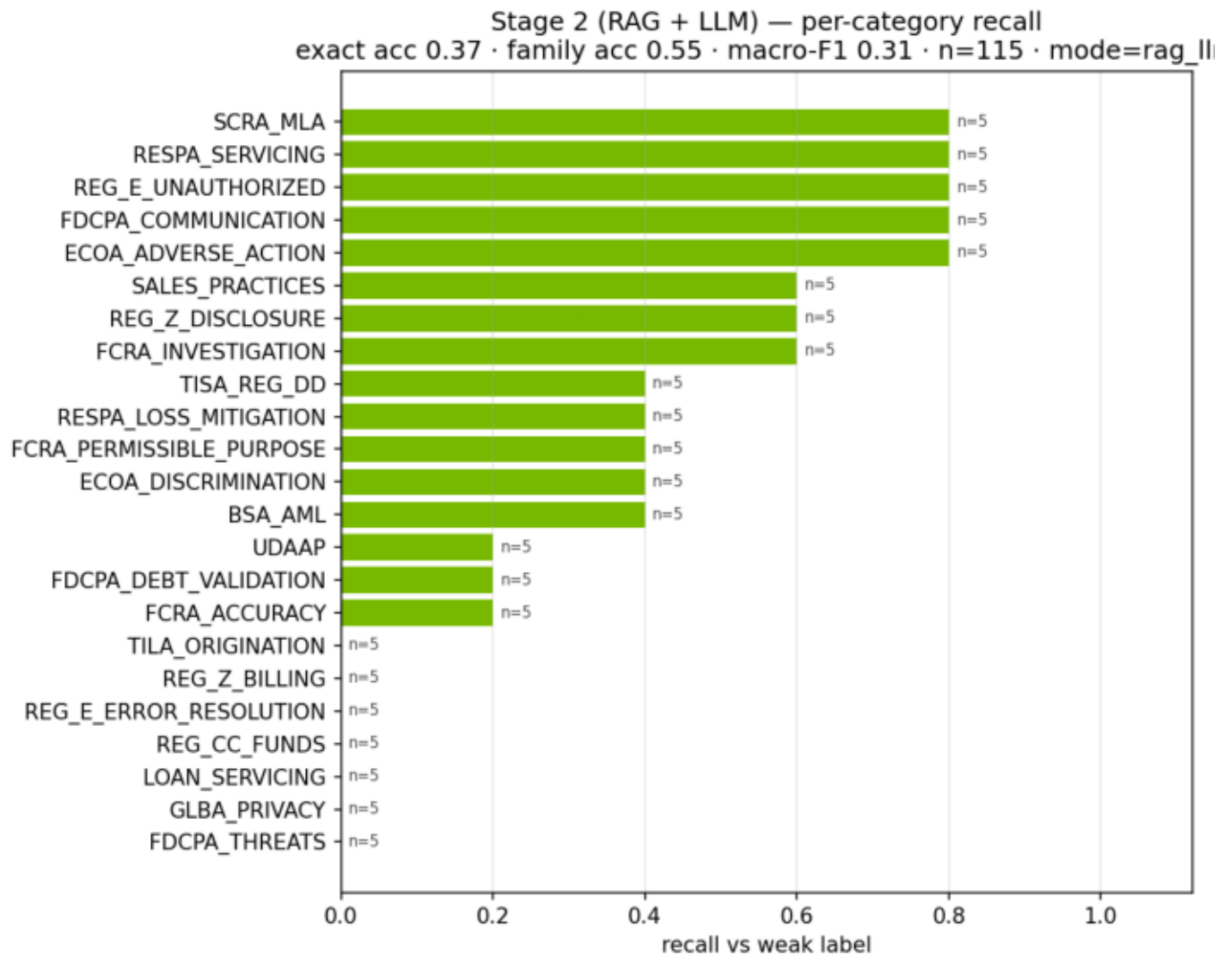


2 · Stage-1 evidence: calibration



Validator re-derived reliability diagram (quantile bins).

3 · Stage-2 evidence



3 • Stage-2 per-category detail

category	support	recall
BSA_AML	5	0.4
ECOA_ADVERSE_ACTION	5	0.8
ECOA_DISCRIMINATION	5	0.4
FCRA_ACCURACY	5	0.2
FCRA_INVESTIGATION	5	0.6
FCRA_PERMISSIBLE_PURPOSE	5	0.4
FDCPA_COMMUNICATION	5	0.8
FDCPA_DEBT_VALIDATION	5	0.2
FDCPA_THREATS	5	0.0
GLBA_PRIVACY	5	0.0
LOAN_SERVICING	5	0.0
REG_CC_FUNDS	5	0.0
REG_E_ERROR_RESOLUTION	5	0.0
REG_E_UNAUTHORIZED	5	0.8
REG_Z_BILLING	5	0.0
REG_Z_DISCLOSURE	5	0.6
RESPA_LOSS_MITIGATION	5	0.4
RESPA_SERVICING	5	0.8
SALES_PRACTICES	5	0.6
SCRA_MLA	5	0.8
TILA_ORIGINATION	5	0.0
TISA_REG_DD	5	0.4
UDAAP	5	0.2

3.1 · Guardrail inventory (validated)

guardrail	mechanism	surfaced as
Taxonomy whitelist	LLM label must be one of the 24 codes; else re	complaint_classifications_total{label}
Strict-JSON parsing	malformed LLM output -> deterministic keyword	mode=fallback counter (alertable spike
Stage-1 gating	non-regulatory volume never reaches the l	cost + attack-surface control
Retrieval grounding	answer must cite a retrieved corpus exce	citation attached to every prediction
Provider portability	OpenA <-> NIM via config; keyword mode if b	mode label per classification

4 · Limitations & disposition

1. Weak labels: ground truth derives from the CFPB issue taxonomy plus keyword rules, not human adjudication. Condition: build a human-adjudicated golden set (≥ 25 per category).
2. Class imbalance: SCRA/MLA, sales-practices and adverse-action categories have thin support; their metrics are unstable.
3. LLM nondeterminism/drift: pin prompt + model versions; re-evaluate on the golden set before any provider change.
4. Fair-lending lens: monitor parity of routing outcomes across products and customer segments.

DISPOSITION: Approve with Conditions. Re-validation triggered by model/prompt change or stage-1 PSI > 0.25 .

Sign-off

Prepared and reviewed under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author, development document: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Author, validation report: Senior Independent Model Validator — PhD (Econometrics), 20 years in second-line validation; independent of the development team per SR 11-7 organizational-independence requirements.